

COMS 4720/5720 Recitation

INTRODUCTION (WEEK 2)

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(Some slides adapted from Dr. Yan-Bin Jia)*

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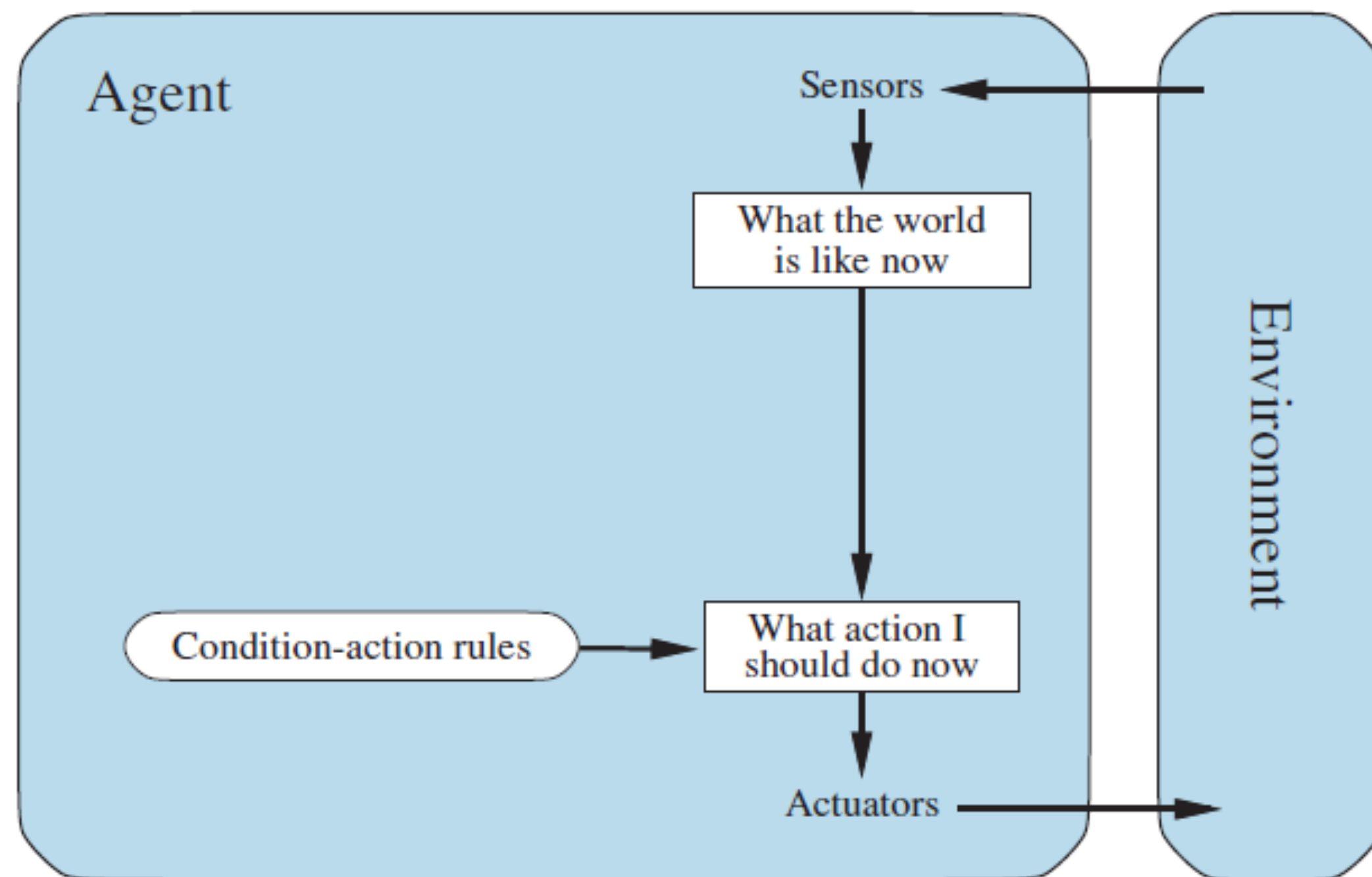
Types of Agents

- Four basic types:
 - Simple Reflex Agent
 - Model-based Reflex Agent
 - Goal-Based Agent
 - Utility-Based Agent

All of them can be converted into:

- Learning-Based Agent

Simple Reflex Agent



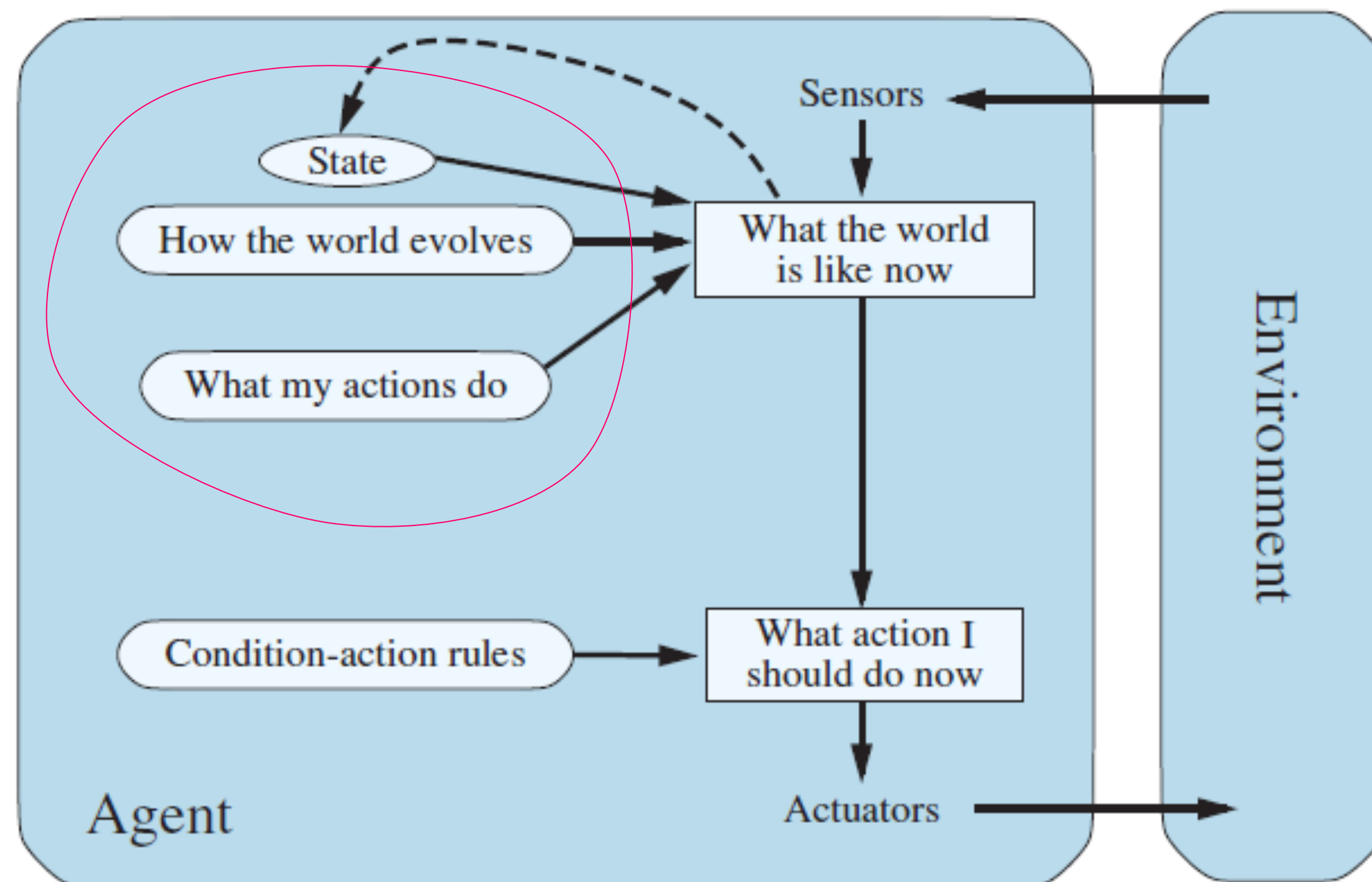
Rectangles: agent's current internal state
Ovals: background information used in the process.

- Select actions based on the current percepts, and ignore the percept history.
E.g., the vacuum agent
- Implemented through *condition-action* rule.

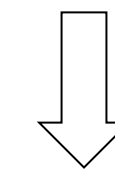
if dirty then suck

if car-in-front-is-braking
then initiate-braking

Model-Based Reflex Agent



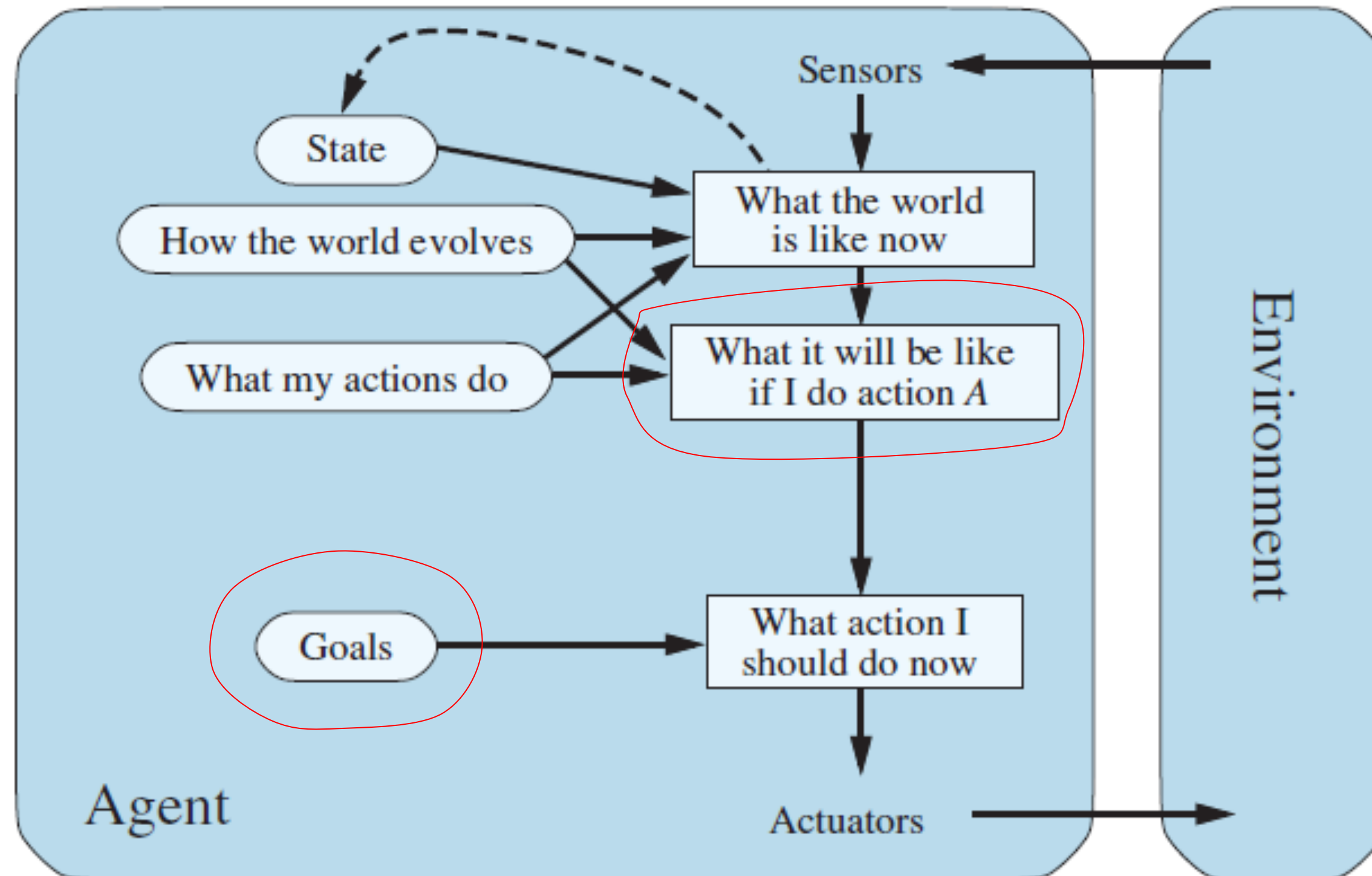
- Partially observable environment.
- Need to maintain some internal state.
- Update it using *knowledge*.



Model of the world

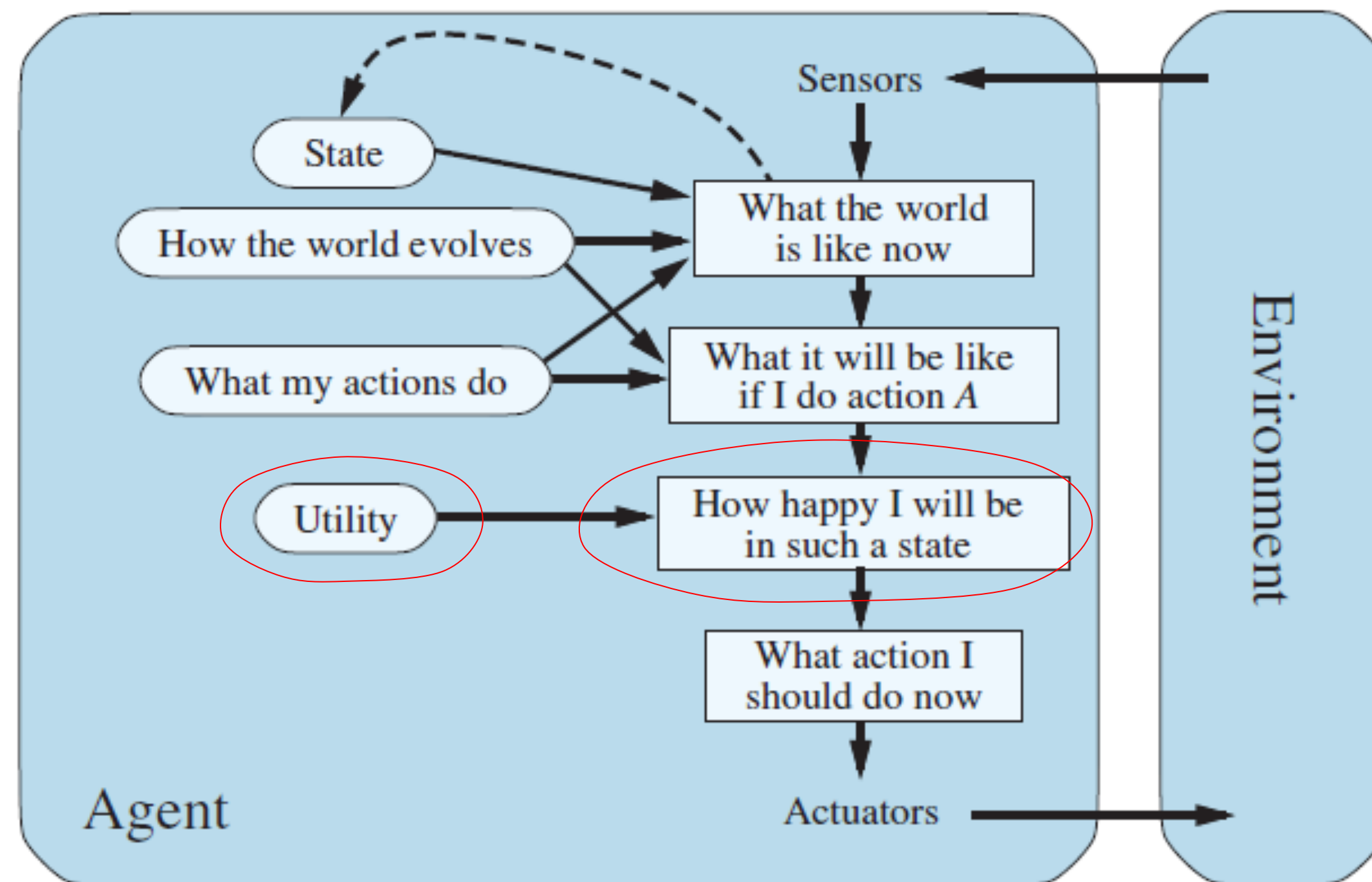
- ◆ Transition model
- ◆ Sensor model

Goal-Based Agent



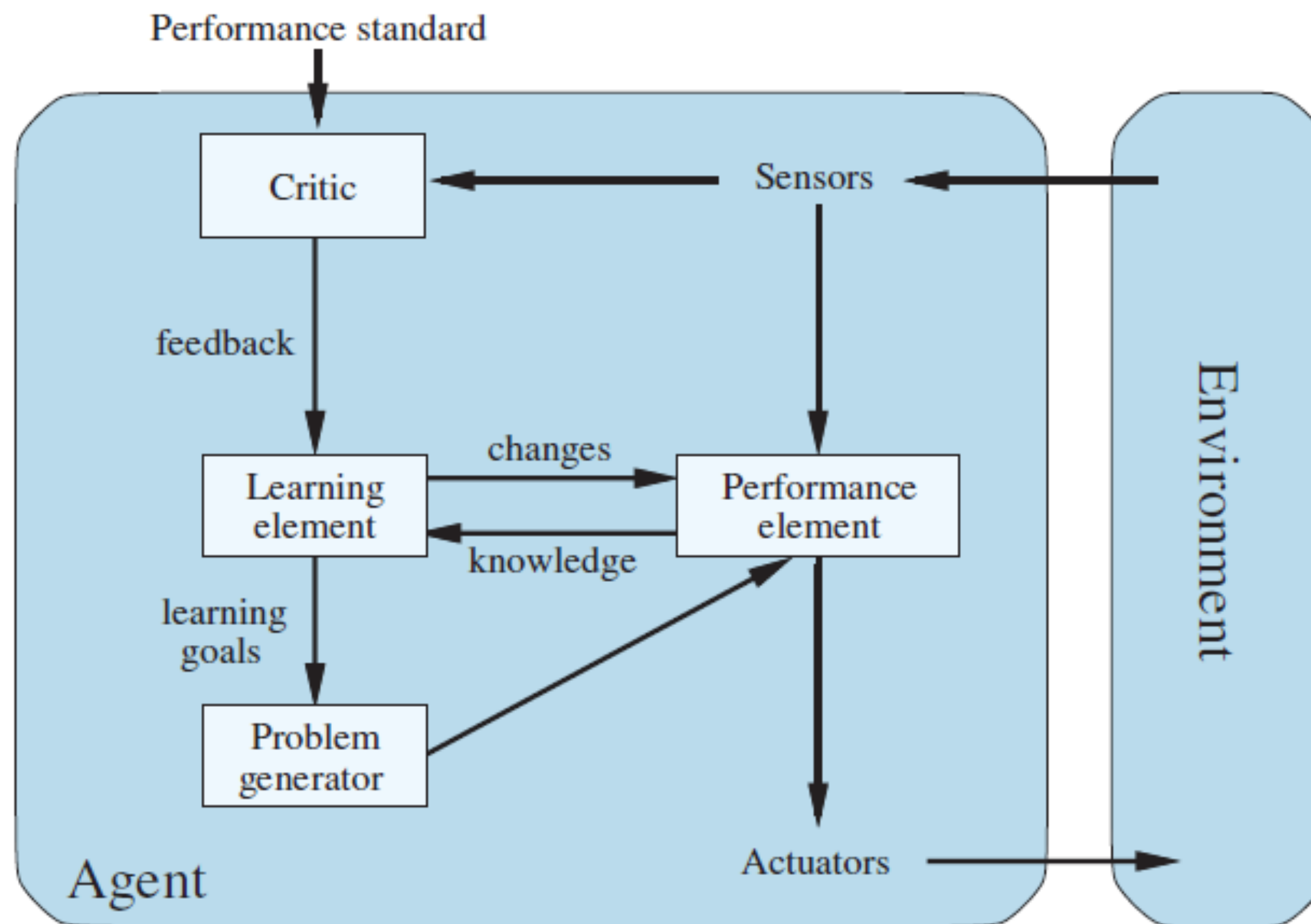
- Needs also some goal information describing desirable situations.
- *Search* and *planning*
when a long sequence of actions is required to find the goal.
- Difference in *taking the future into account*.

Utility-Based Agent



- Different ways to achieve a goal sometimes.
- Use a *utility function* that maps a sequence of states to a real number *utility*
↑
internal performance measure
- Maximize expected utility.

Learning-Based Agent



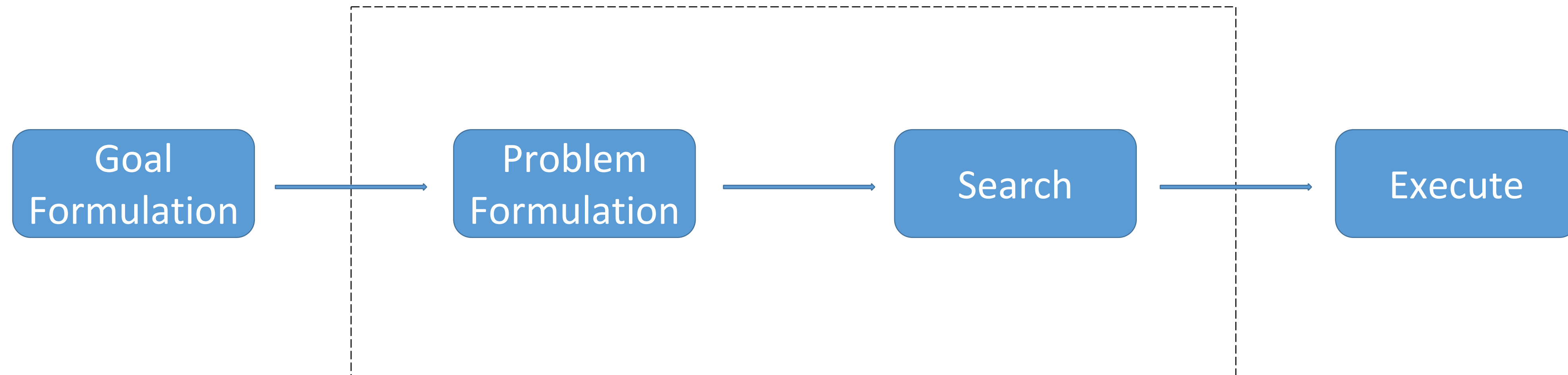
- *Critic* provides feedback on the agent's performance based on fixed performance standard.
- *Learning element* introduces improvements in performance element.
- *Performance element* selects actions based on the percepts.
- *Problem generator* suggests actions that will lead to new and informative experiences.

Performance measure

- An **omniscient** agent knows the actual outcome of its actions.
Impossible in reality!
- **Rationality** maximizes the **expected** performance.
- **Perfection** maximizes **actual** performance.

Rationality $\neq$ omniscience $\neq$ perfection

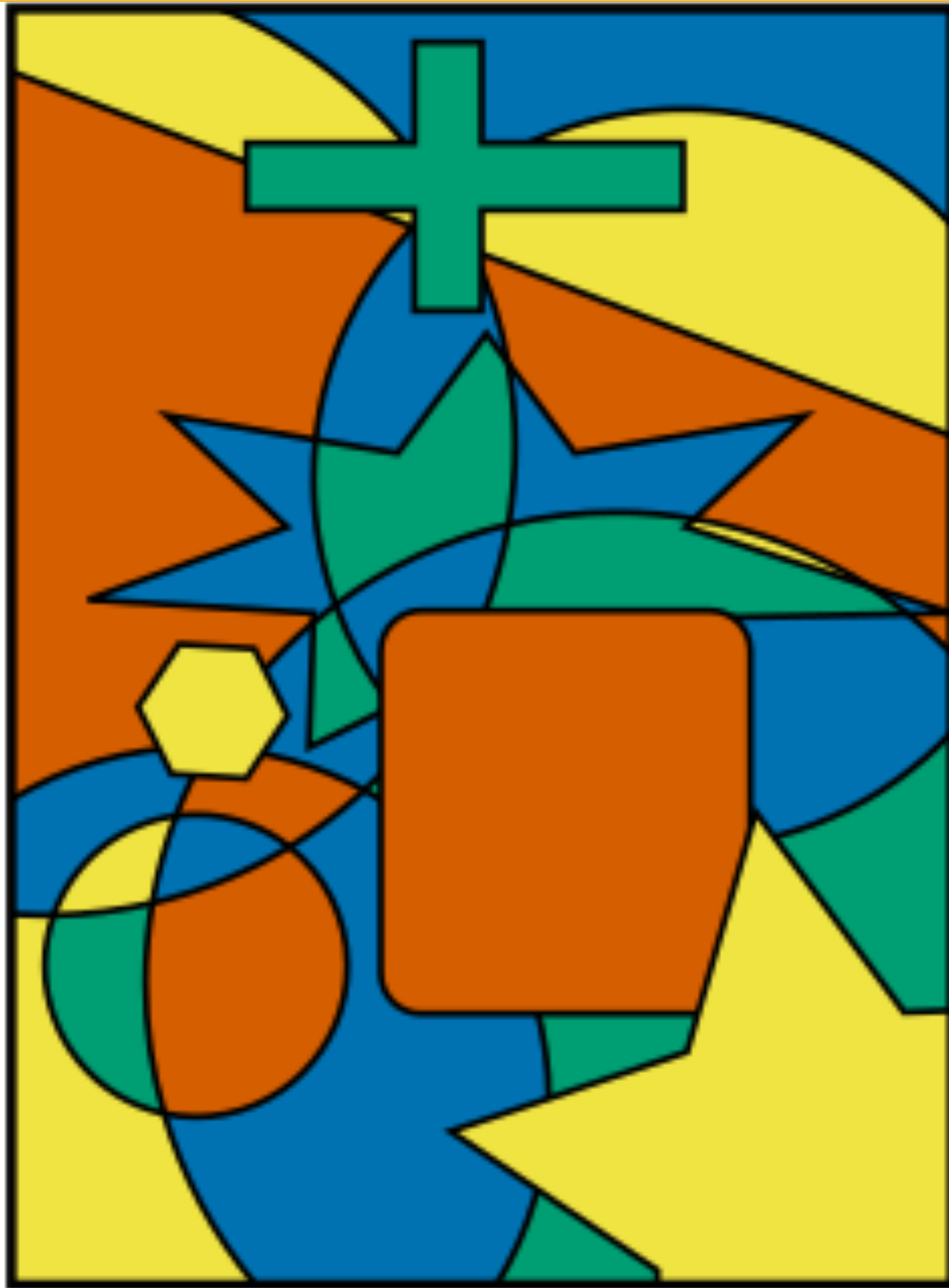
Problem Solving Process



Problem Formulation

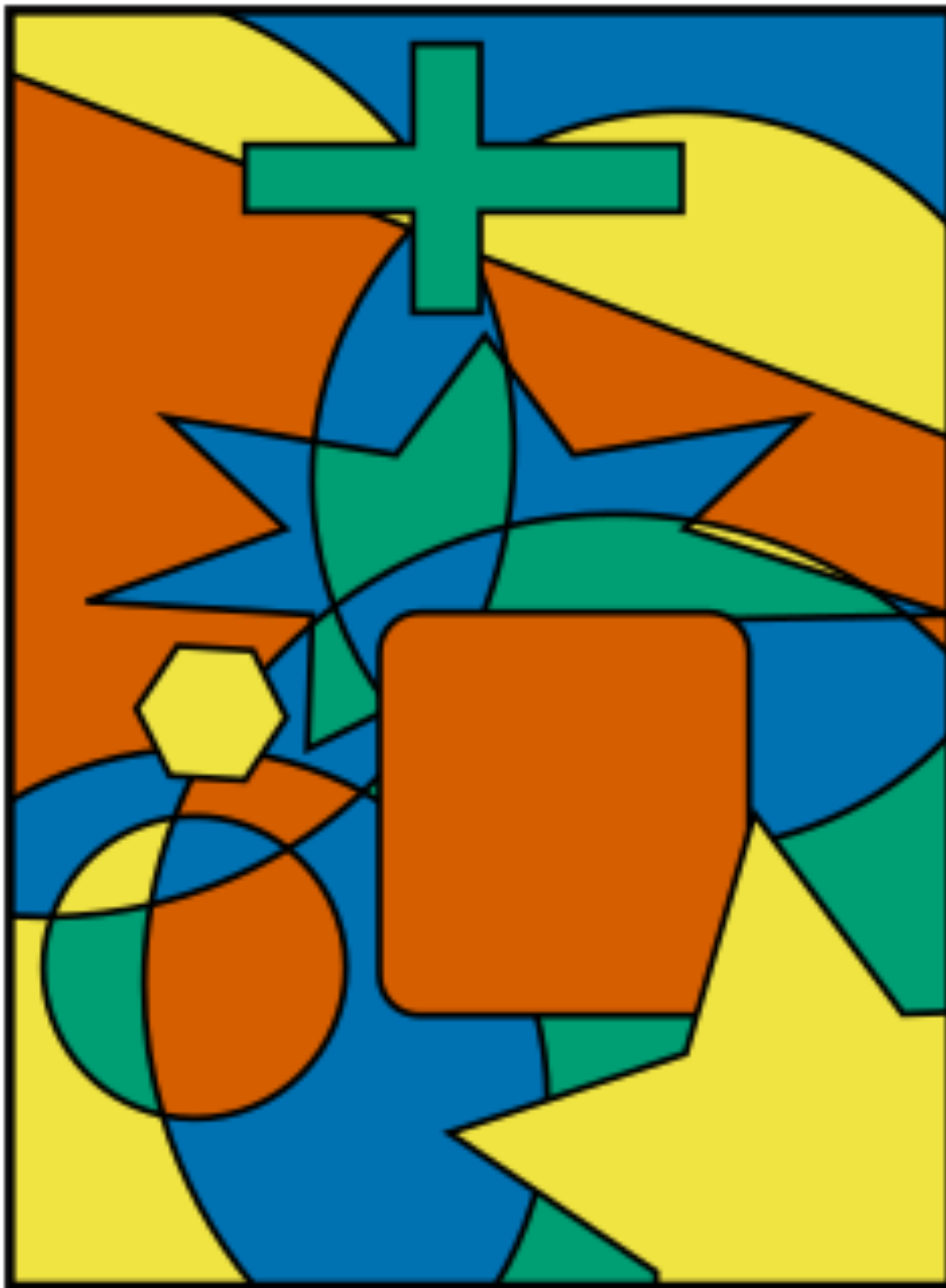
- 1. States**
- 2. Initial State**
- 3. Goal State**
- 4. Actions**
- 5. Transition model**
- 6. Cost**

Example 1: Four Color Problem



Using only four colors, you have to color a planar map in such a way that no two adjacent regions have the same color.

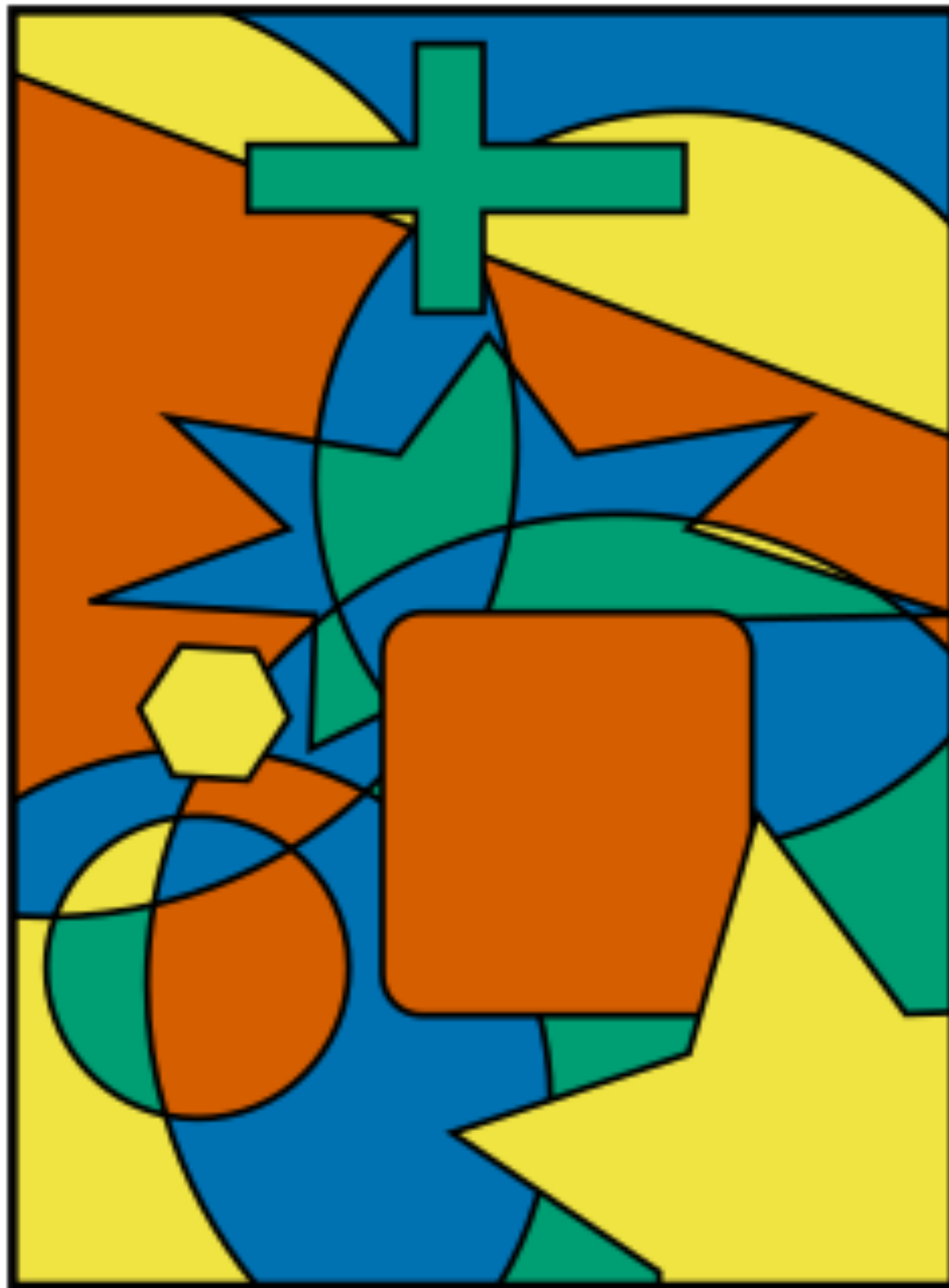
Example 1: Four Color Problem



States: Each state is a tuple representing the color of each region, with possible values of 0 (uncolored), 1, 2, 3, or 4. If there are n regions, then 5^n states are possible, as each region can be in one of 5 states (0 to 4).

Initial State: All regions are uncolored.

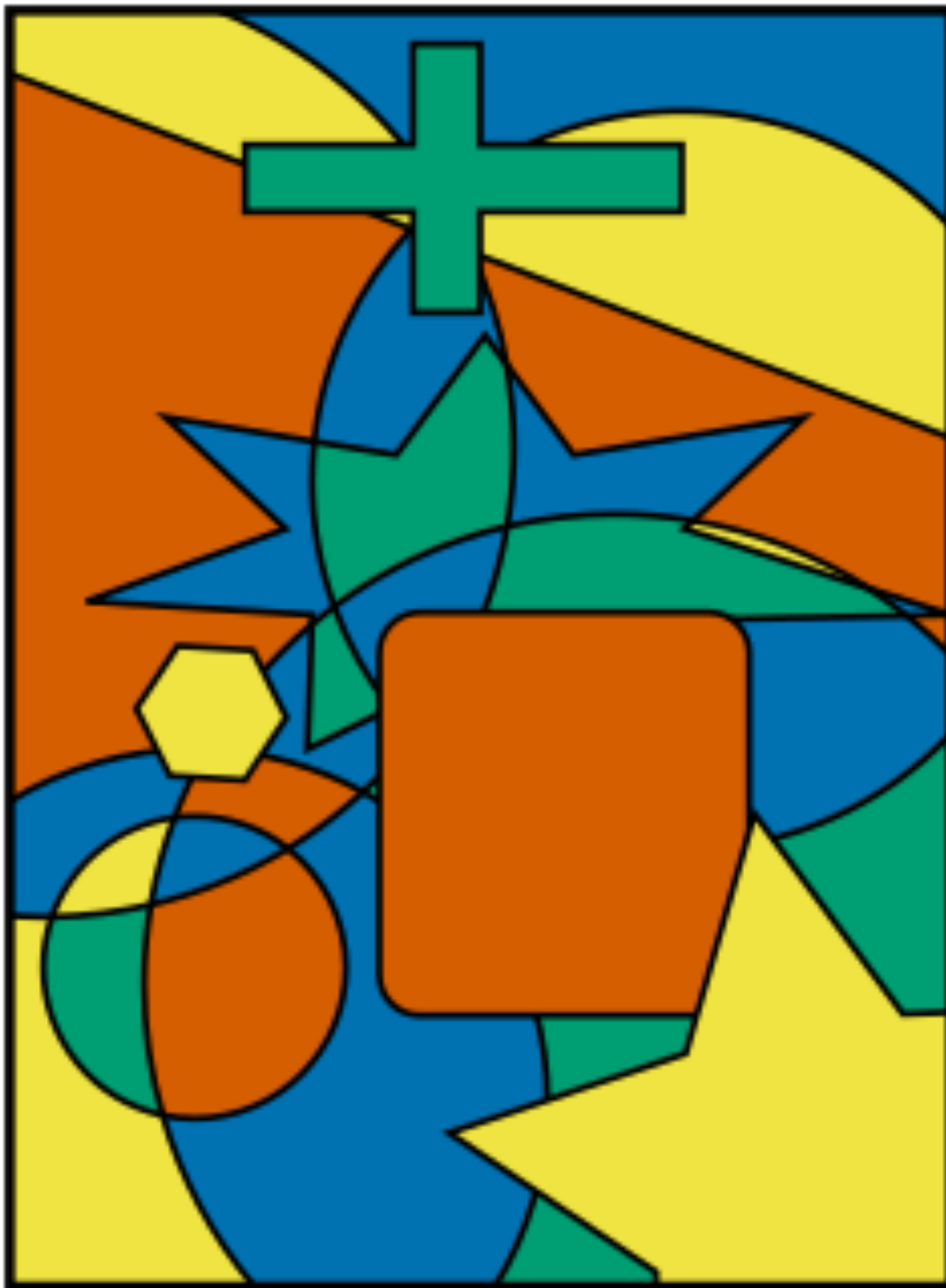
Example 1: Four Color Problem



Actions: *Assign one of four colors to an uncolored region*

Transition Model: *If a region is uncolored, color it ensuring no two adjacent regions share the same color.*

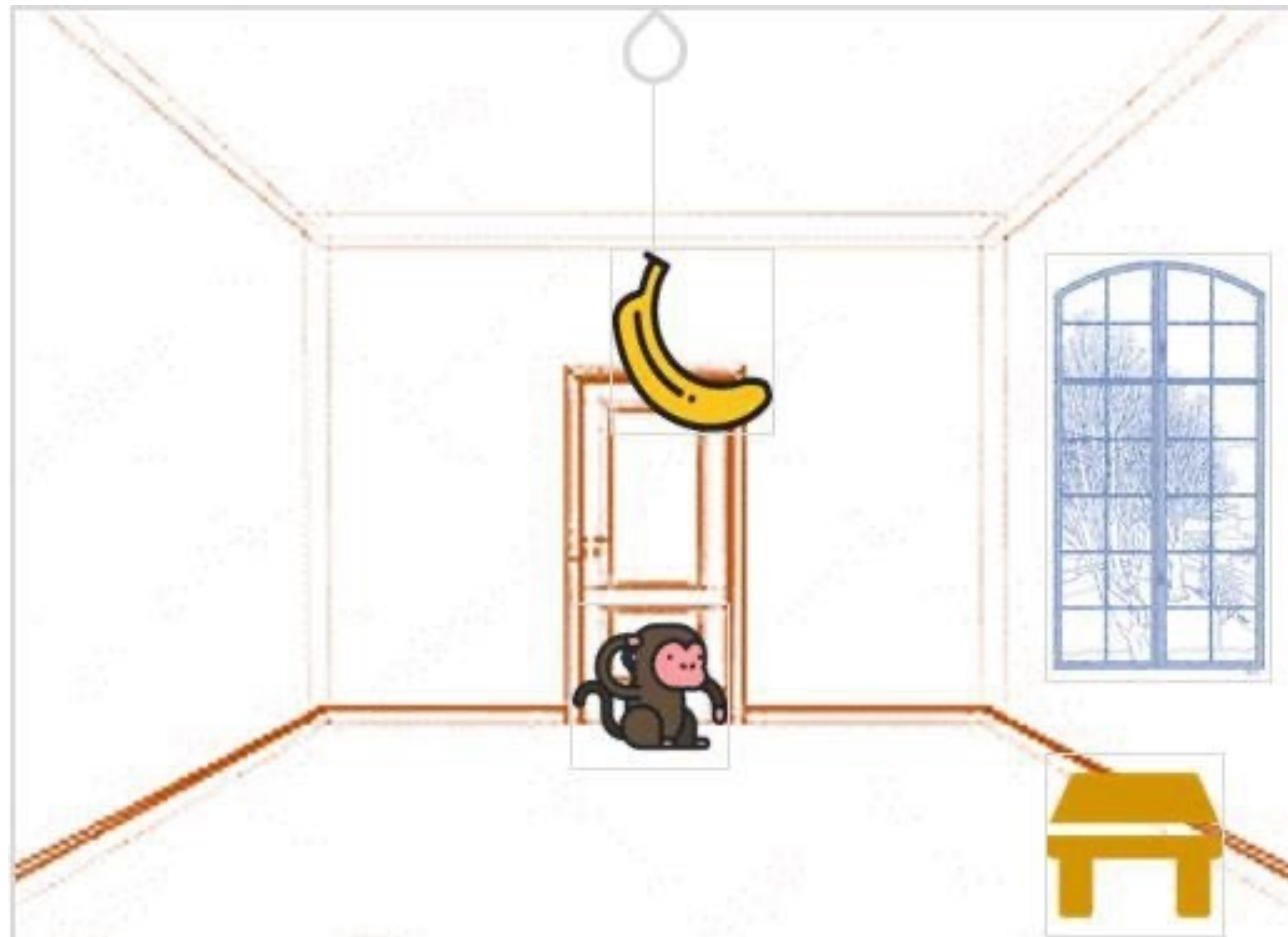
Example 1: Four Color Problem



Goal State: All regions are colored, no two adjacent regions share the same color.

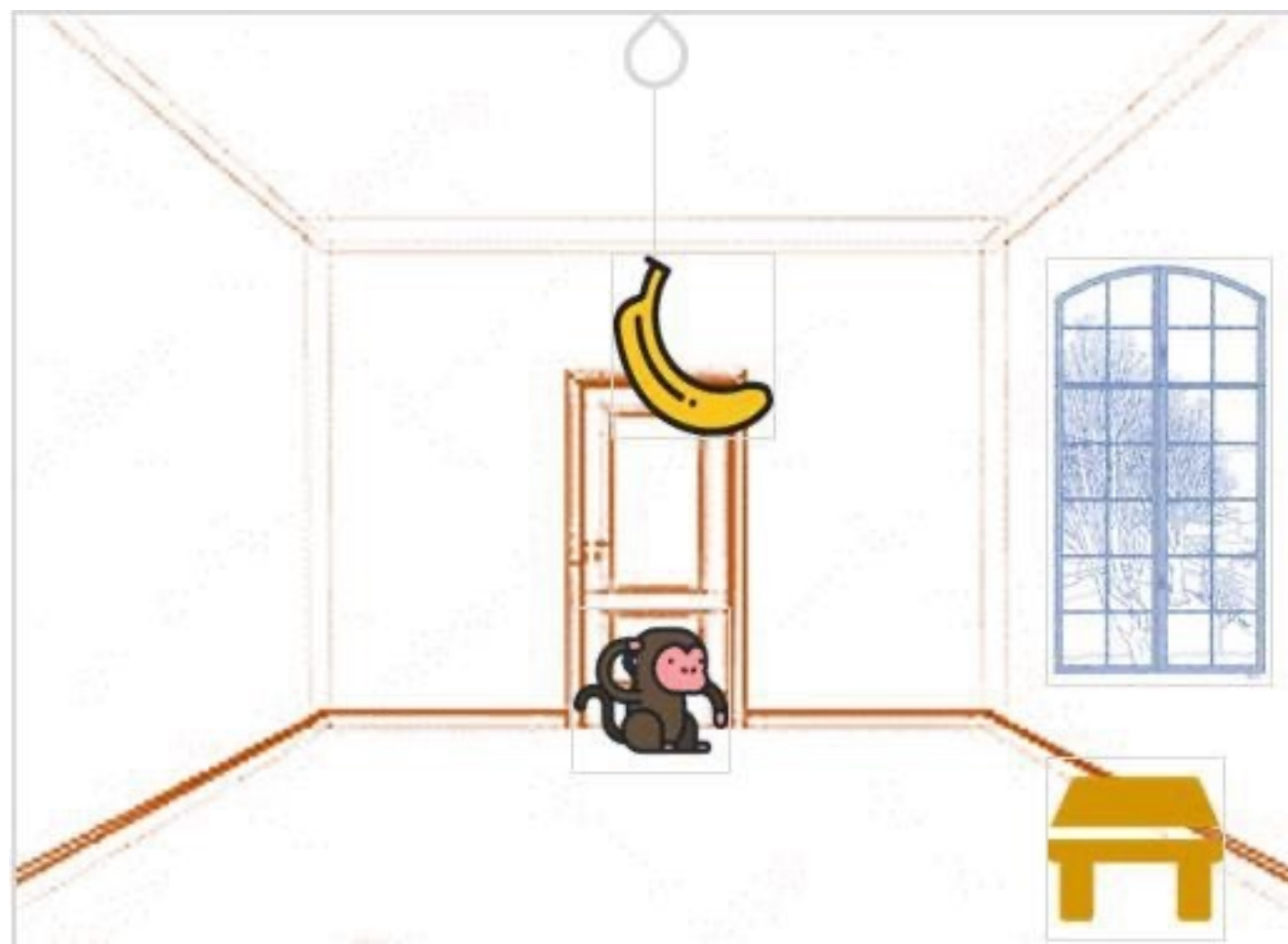
Cost: Each coloring action incurs a cost of 1.

Example 2: Monkey Problem



3-foot-tall monkey is in a room where some bananas are suspended from the 8-foot ceiling. He would like to get the bananas. The room contains two stackable, movable, climbable 3-foot-high crates

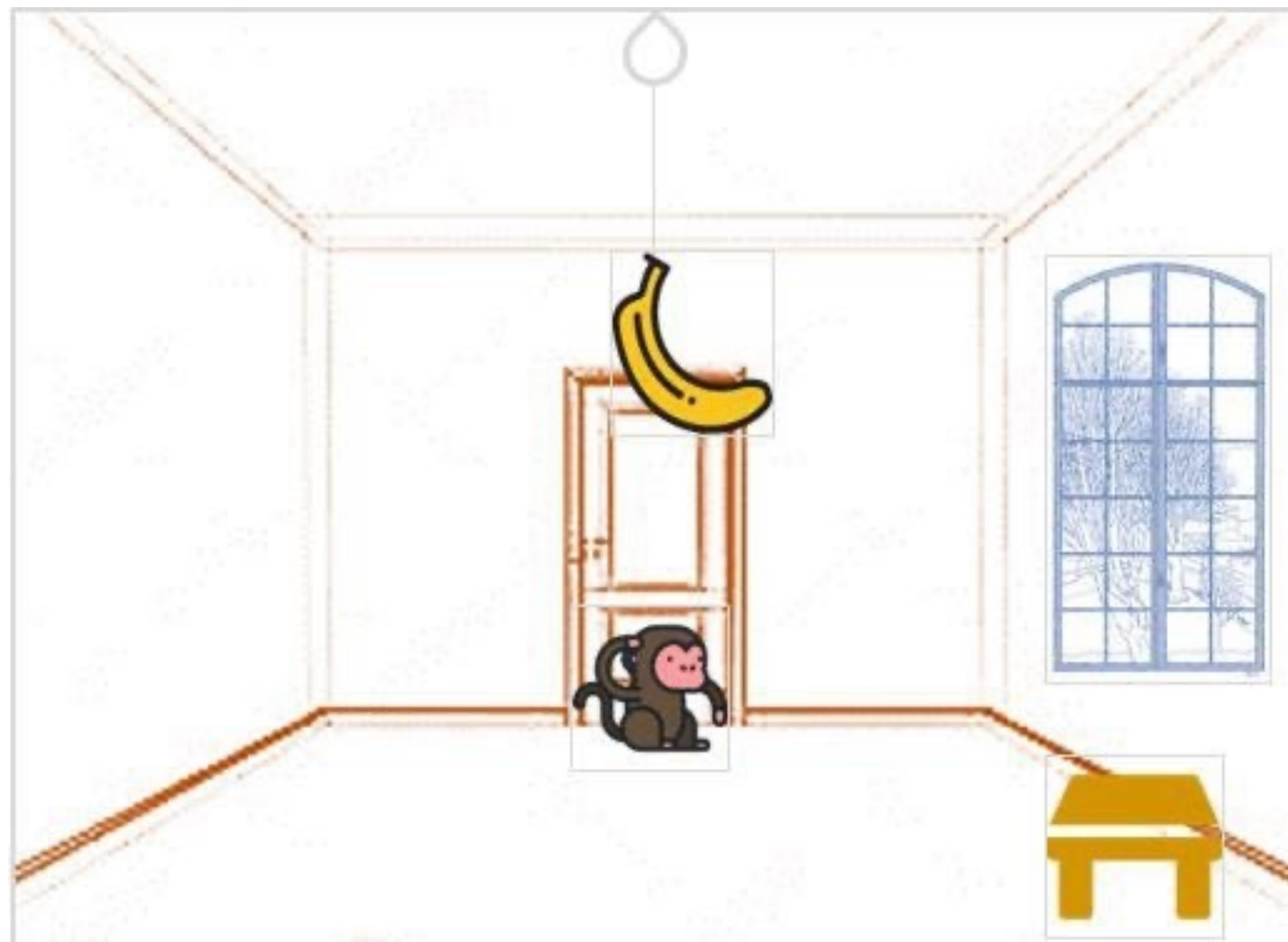
Example 2: Monkey Problem



States: *state is a tuple showing the monkey's position, the crates' configuration, and the banana's position.*

Initial State: *Monkey on the floor, crates unstacked and separate, bananas at the ceiling.*

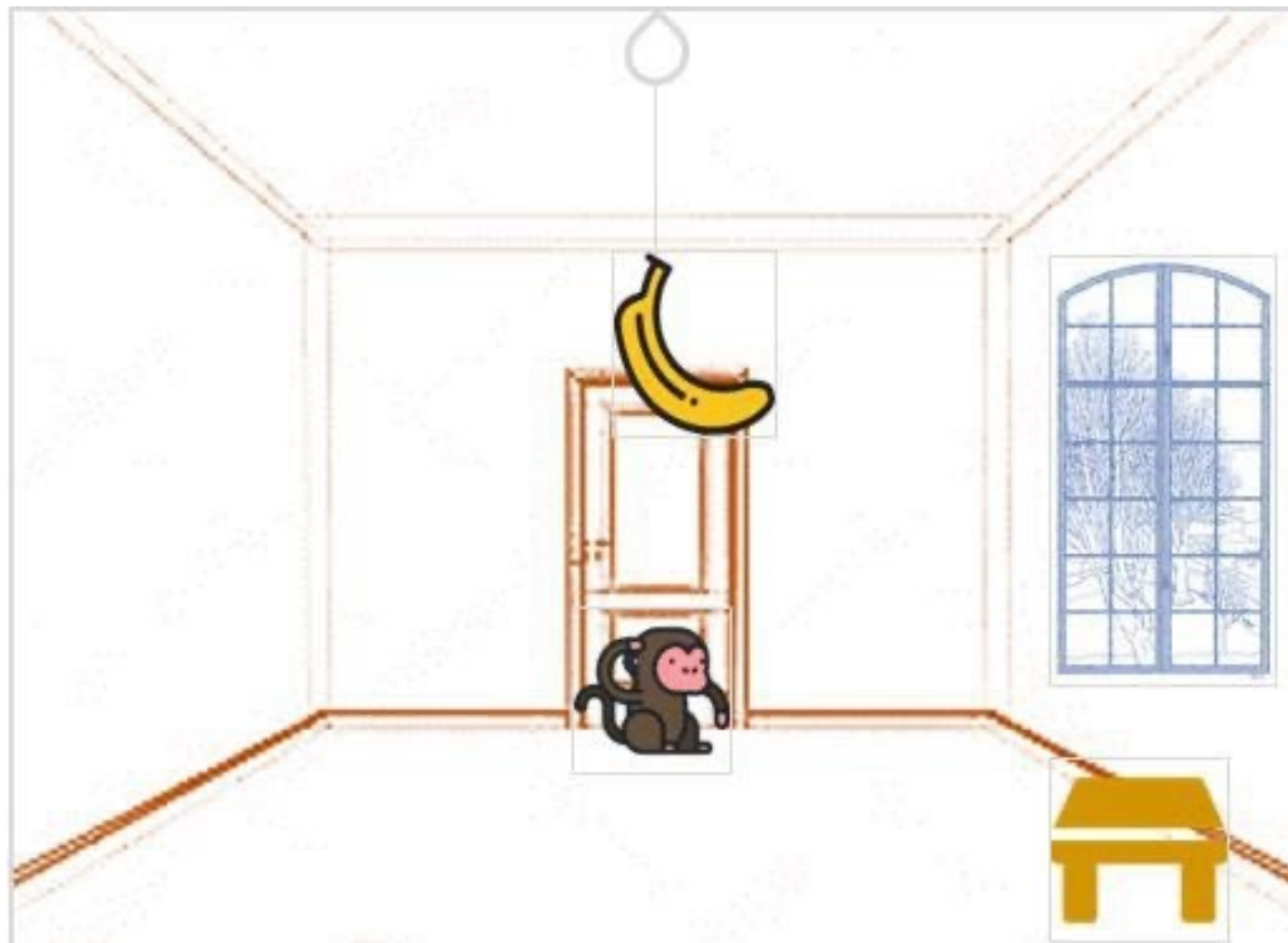
Example 2: Monkey Problem



Actions: Monkey moves, climbs, or stacks crates.

Transition Model: Update positions of crates and monkey's position based on actions taken

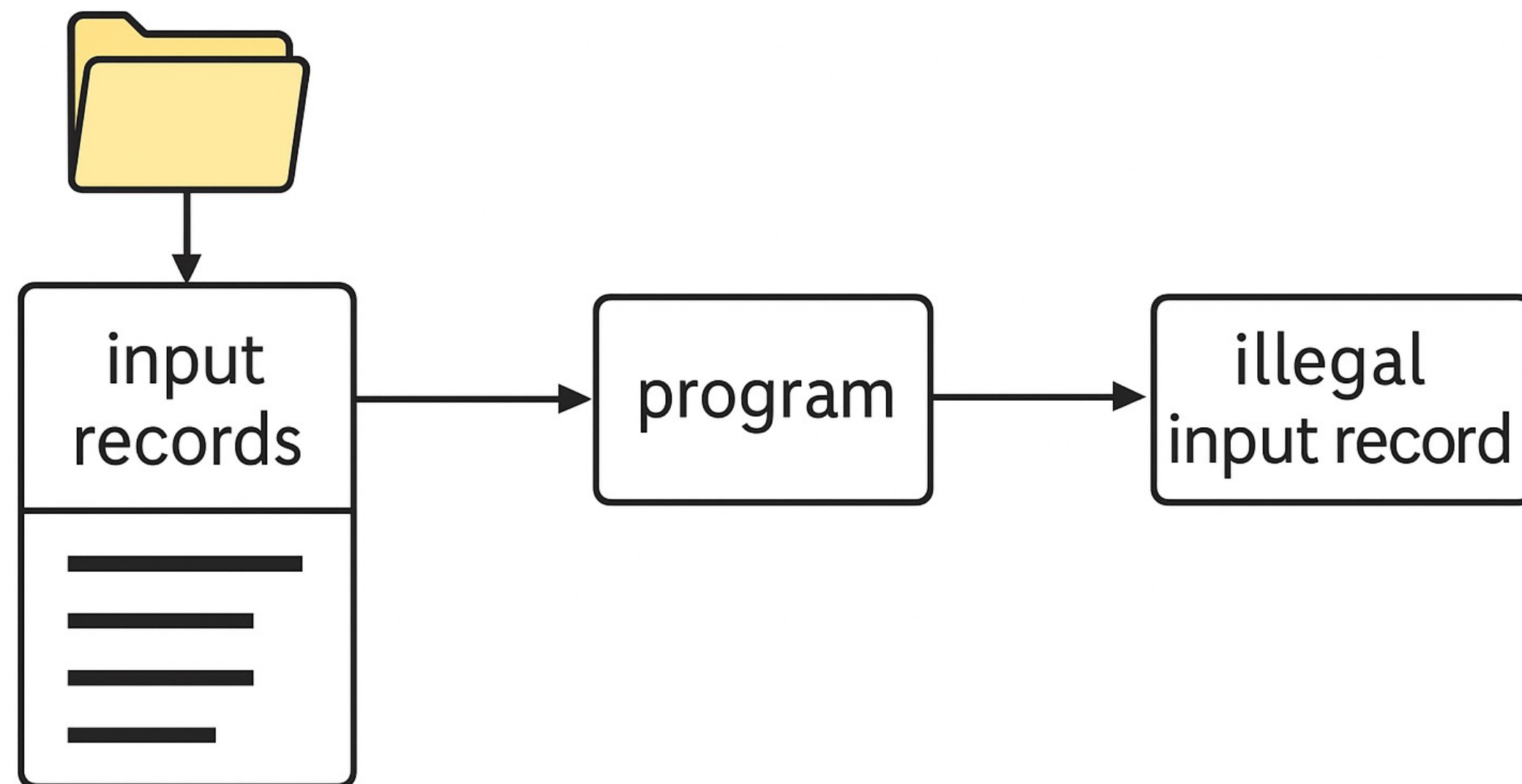
Example 2: Monkey Problem



Goal State: *Monkey has bananas.*

Cost: *Each Costs 1, total cost is the number of actions*

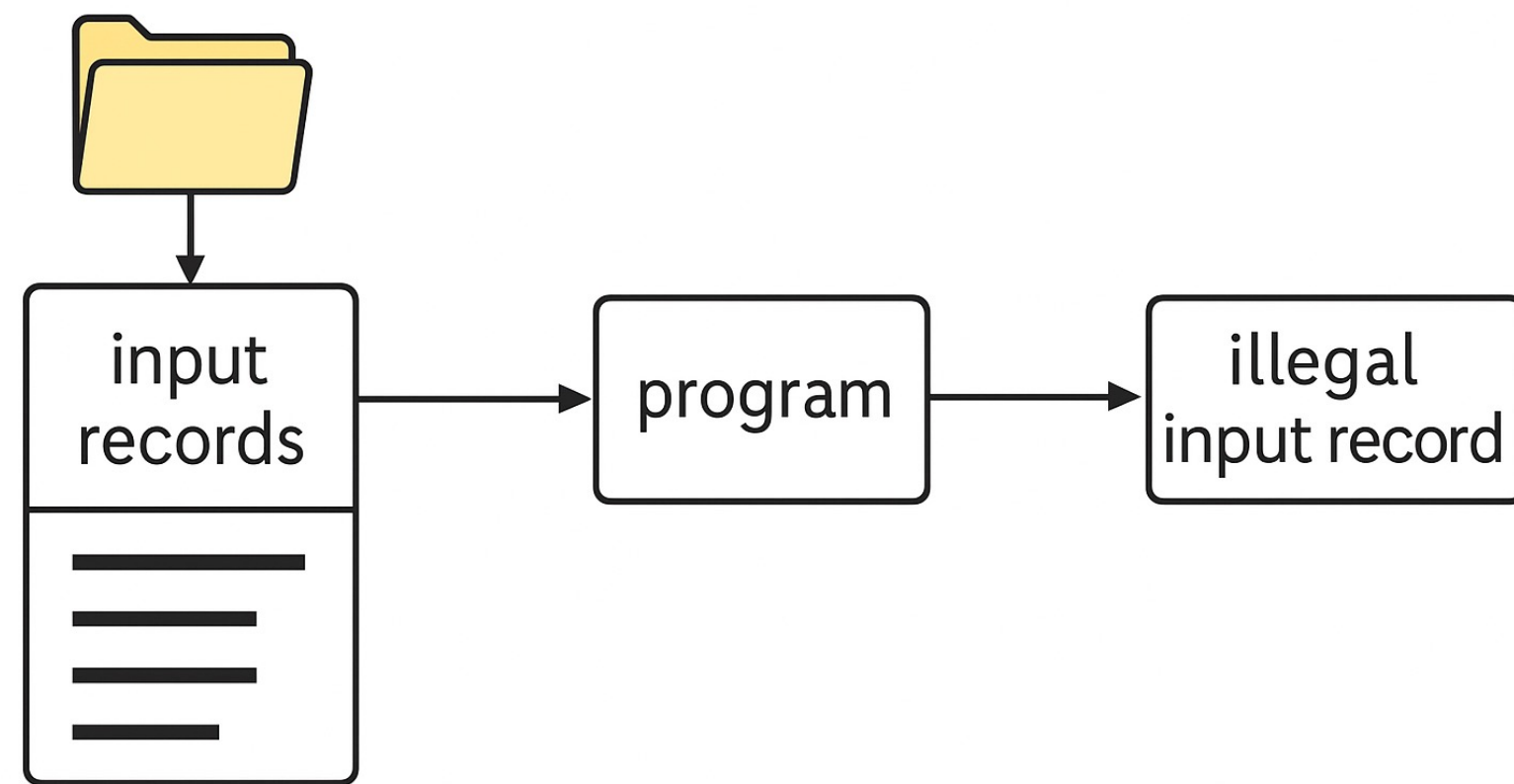
Example 3: illegal record Problem



You have a program that outputs the message “illegal input record” when fed a certain file of input records. You know that processing of each record is independent of the other records. You want to discover what record is illegal

To discover the illegal record

Example 3: illegal record Problem

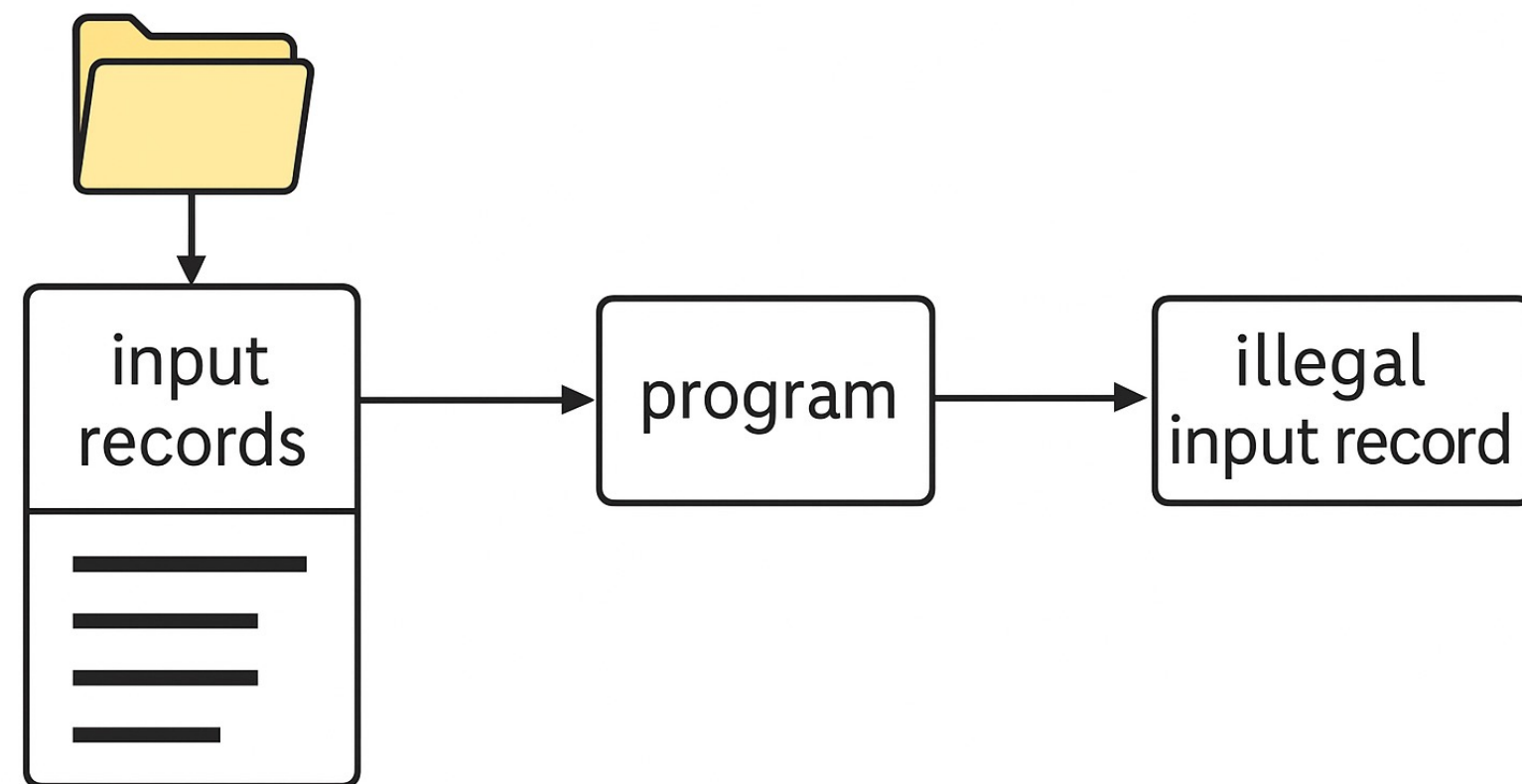


To discover the illegal record

States: A state is represented as a tuple of n elements, $s_1, s_2 \dots s_n$ where each s_i can be either 0 (unprocessed) or 1 (processed). This representation captures the processing status of each record in the file. If there are n records, then 2^n states are possible (each record can be either processed or unprocessed).

Initial State: All records are unprocessed.

Example 3: illegal record Problem

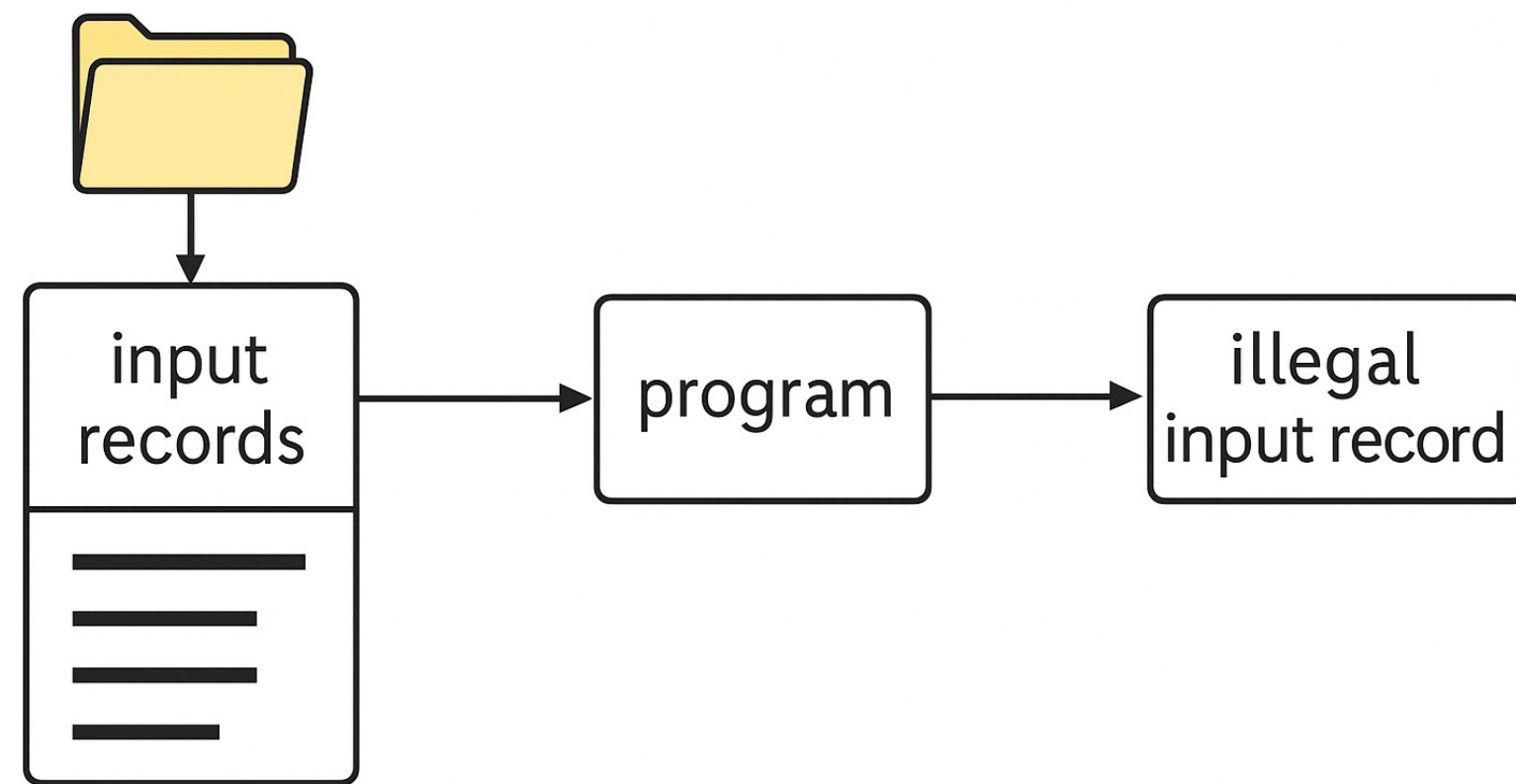


Actions: Process a record.

Transition Model: Change a record's state from unprocessed to processed.

To discover the illegal record

Example 3: illegal record Problem



To discover the illegal record

Goal State: *Identify and isolate the illegal record(s).*

Cost: *Each Costs 1, total cost is the number of actions*

Example 4: Filling Jug Problem

You have a 12-gallon, an 8-gallon, and a 3-gallon jug. Measure out exactly 1 gallon.



You have three jugs, measuring 12 gallons, 8 gallons, and 3 gallons, and a water faucet. You can fill the jugs up or empty them out from one to another or onto the ground. You need to measure out exactly one gallon.

Example 4: Filling Jug Problem

You have a 12-gallon, an 8-gallon, and a 3-gallon jug. Measure out exactly 1 gallon.



States: The current amount of water in each of the three jugs, represented as a tuple (x, y, z) . Since the jugs have capacities of 12, 8, and 3 gallons, there are $13 \times 9 \times 4 = 468$ possible states.

Initial State: All jugs are empty

Example 4: Filling Jug Problem

You have a 12-gallon, an 8-gallon, and a 3-gallon jug. Measure out exactly 1 gallon.



Actions: Fill a jug, empty a jug, or transfer water between jugs.

Transition Model: Adjust the amounts in the jugs based on the action taken.

Example 4: Filling Jug Problem

You have a 12-gallon, an 8-gallon, and a 3-gallon jug. Measure out exactly 1 gallon.



Goal State: *Any jug contains exactly 1 gallon.*

Cost: *Each Costs 1, total cost is the number of actions*



That's all Folks!